

Research paper

Novel deep learning framework for rainfall forecasting integrating generative adversarial and spatiotemporal graph neural networks

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ABSTRACT

Global warming and climate change cause intense extreme precipitation, demanding advanced rainfall prediction techniques for improved water resource planning and management. This study developed two deep-learning algorithms, Spatio-Temporal Graph Neural Networks (ST-GNNs) and Generative Adversarial Graph Neural Networks (GA-GNNs), and evaluates their performance on the entire meteorological data of Thailand in the years 1993 to 2022. The findings indicate that ST-GNNs outperform GA-GNNs in terms of accuracy: ST-GNN has a high coefficient of determination ($R^2 = 0.9580$), a low root mean square error (RMSE = 0.1560), and mean absolute error (MAE = 0.3950), which proves its ability to capture complex spatiotemporal rainfall patterns. In contrast, GA-GNN has poor performance measures ($R^2 = 0.5800$, RMSE = 0.8400, MAE = 1.5770) that confirm the fact that it has little representational capabilities over the intricate rainfall dynamics. SHAP analysis has shown that wind speed, relative humidity, and minimum temperature have the most significant impact on ST-GNN predictions, and wind speed is the most positive factor (SHAP value = 0.634). Moreover, the paper records that ST-GNNs are superior when it comes to rainfall category classification, with a weighted F1 score of 1.00 and an accuracy of 0.992- a difference that is much better than that of GA-GNNs. The results highlight the importance of considering both spatial and temporal dependencies in modeling rainfall in climatically complex environments like Thailand.

1. Introduction

Weather and climate play a primary role in influencing public health outcomes, economic activities, ecosystem stability, and governance at a global scale [1]. Their variability and long-term changes influence societal resilience, policy frameworks, and sustainable development strategies [2,3]. Weather forecasting focuses on short-term atmospheric conditions, while climate modeling deals with long-term changes in atmospheric variables [4]. Accurate and earlier forecasting of extreme weather events and other significant climate shifts demand have increased significantly with the frequency of global warming and climate change [5–7]. Rainfall prediction is a critical challenge for meteorologists as it is key to monitoring adverse weather events, and precipitation data is fundamental to meteorological systems [8]. Rainfall can influence social and economic systems over a wide range [9].

Disasters such as floods [10], landslides, and mudflows [11] caused by extreme rainfall events pose severe threats to human lives and assets [9]. Although primarily used, traditional numerical weather prediction (NWP) models are often too approximate due to inadequacy in the approximation of dynamical and physical processes that define the weather phenomena [8]. To overcome these limitations, statistical models have been used to post-process NWP outputs or directly forecast precipitation [12,13]. There is promise in these models to do away with some of the inherent flaws in NWP models [8].

In recent years, artificial intelligence (AI), including machine learning (ML) and deep learning (DL) techniques, has attracted much attention because its AI models based on temporal dependencies of rainfall and methods such as Autoregressive Integrated Moving Average (ARIMA) [14,15], Classification and Regression Trees (CART) [16], and Long Short-Term Memory (LSTM) networks [17] were partially

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effective. However, these models offer good results but are limited by the temporal dependence they pay attention to at the expense of addressing the spatial relations essential to predicting precipitation [18]. Additionally, a single AI model cannot fully explain the processes that generate rainfall [8]. Past studies showed the ability of traditional ML algorithms, i.e., multiple linear regression [19] and support vector machines [6], to predict the weather. However, recent advancements in DL methods, including convolutional neural networks (CNNs) [20] and recurrent neural networks (RNNs) [20], have demonstrated increasing accuracy for forecasting purposes.

Among advanced DL methods, Graph Neural Networks (GNNs) proved to be a promising method for weather forecasting [21]. GNNs are graph-structured data, nodes are entities, and edges are relationships [22]. With its locality and structure exploitation, GNNs are the core strength of GNNs to accurately model the interactions between different regions and periods [22,23]. Data presented as graphs have been successful with GNNs, and GNNs are already being used in problems such as weather prediction [24]. GNNs can find complex relations between meteorological variables and learn from a big dataset, which is ideal for prediction [25]. Since then, the advent of Dynamic GNNs is also capable of working with dynamic (time series) graph structures, which has further supported the potential of GNNs in weather forecasting [26].

Furthermore, Wu et al take up GNNs to model the spatial relationship among rain gauges in Eastern China [23], and Yang et al show that GNNs can also be utilized to model interactions among observation stations during heatwave events [27]. In traditional GNNs, the problems arise from handling spatial and temporal dependencies in specific data [28]. As these models prefer the temporal evolution of spatial phenomena, they are perfectly suitable for the temporal dimension of the rainfall prediction problem in which space and time are of utmost concern for the dynamics of weather systems [29]. Spatio-temporal GNNs (ST-GNNs) have recently been shown to be effective for environmental modeling by predicting rainfall patterns, which is modeling spatial correlations across different geographical regions while considering the temporal change of such correlations [30]. By considering historical weather data and spatial relationships among regions, ST-GNNs can get more accurate and robust predictions [31]. Jiang et al introduce CSTGNN-a CausalST-GNN, or a Causal Spatial-Temporal Graph Neural Network, which combines Graph Neural Networks and Gated Recurrent Units- to optimize the prediction of storm surges throughout coastal observation stations. CSTGNN trained on 13 years of in situ observations in Florida outperforms six other benchmarks in the ability to capture spatial dependencies and coastal terrain detail [32]. A similar research effort leads to the so-called TC-diffusion-based generative model of the complete simulation of tropical cyclone trajectories that promotes consistency with past observations and completely reflects spatial heterogeneity and prior information. The model is also effective in the evaluation of wind hazards in the southeastern coastal region of China [33]. Generative Adversarial Networks, as well, have been discovered as a potential way of boosting accuracy in complicated forecasting areas. The example of one research path shows this potential in the form of better precipitation nowcasting: integrating adversarial regularization in radar echo forecasting with ConvRNNs and using multiscale WGAN, researchers attain better forecast results than other available options, producing more accurate and realistic results [34].

In Thailand, the combination of limited historical climate data, monsoon patterns, geographical features, and climate variability makes precipitation forecasting difficult [35]. The complexity of rainfall prediction in the tropical climate of Thailand stems from the region's highly variable climate, influenced by monsoons and seasonal shifts [20,35,36]. While traditional approaches, including the NWP model and ML techniques like LSTMs and XGBoost, have shown effectiveness, the limitation in capturing spatiotemporal dependency and extreme precipitation events remained unsolved. Another study improved prediction accuracy with spatial correlations in meteorological data using GNNs and adversarial variants like GA-GNNs. RMSE and NSE cannot

only be improved, but peak flow overestimation remains like their predecessors, AEMP-GAN [37], LSTM-based approaches [38], or wavelet-transformed LSTM models [20] that had promising results but failed to generalize well for hidden precipitation patterns. The Wasserstein GAN (WGAN) model efficiently reproduces geostrophic balance but was less successful with small scales [39]. Furthermore, the past GNNs and other hybrid DL-based related work is summarized in Table 1. The current prediction models and AI techniques being used in Thailand have limitations [8,18].

The proposed research combines ST-GNNs and GA-GANs to advance rainfall prediction accuracy levels. The research combines GNNs and GANs through a generator-discriminator structure, improving the ability to extract sophisticated data distribution patterns. GA-GANs learn detailed rainfall patterns through adversarial training, effectively generating authentic spatiotemporal rainfall distributions as reported in [40,41]. Research has shown that ST-GNNs produced strong representations of rainfall patterns by processing spatial patterns between locations and temporal progression over time [23,27,28,31,42]. The combination of ST-GNNs and GA-GNNs demonstrates the potential to enhance rainfall prediction precision. The challenges raised in this work highlight the intrinsic need for ST-GNNs and GA-GNNs with spatial and temporal dynamics to achieve a more robust prediction for rainfall. To address the limitations of previous models, these advanced DL models are developed and implemented in this study to achieve better generalization, higher accuracy, and better capability in extreme rainfall event prediction for Thailand's unique climate conditions.

The main goals of this study are: (1) development and evaluation of ST-GNNs and GA-GNNs for improving rainfall prediction accuracy. (2) To compare the predictive performance of ST-GNNs and GA-GNNs in capturing spatio-temporal rainfall dependencies in Thailand. This paper is structured as follows: Section 2 describes the dataset, outlines the methodology and model construction, Section 3 presents the experimental results, and Section 4 concludes with discussions and future research directions.

2. Materials and methods

2.1. Study area and data collection

Thailand, situated between latitudes 5°37' to 20°27' N and longitudes 97°22' to 105°37' E (Fig. 1), is classified into five distinct climatic zones by the Thai Meteorological Department (TMD): central, eastern, north-eastern, northern, and southern regions. The country experiences a tropical climate with consistently high yearly temperatures [36]. The southwest monsoon, which occurs from mid-May to mid-October, transports warm, moist air from the Indian Ocean, resulting in the rainy season with peak rainfall in July and August. In contrast, the winter season in northern Thailand, lasting from mid-October to mid-February is influenced by the northeast monsoon, bringing cool, dry air from China [35]. The summer season, from mid-February to mid-May, is marked by increasing temperatures and transitional weather patterns, with April being the hottest month before shifting to the southwest monsoon [46].

The dataset spanning the past 30 years (1993–2022), including daily measurements of rainfall, relative humidity, wind speed, wind direction, and maximum and minimum temperatures, was collected from 129 TMD stations. Numerous missing values were identified within the dataset. To address this issue and ensure the data's quality, a thorough quality control procedure was conducted, and all missing values were imputed using the method proposed by Wangwongchai et al [47]. Stations installed after 2015 were excluded from the imputation process due to >50 % missing data across all variables. Fig. 2 provides detailed statistics of the dataset.

Table 1

Summary of past work on rainfall predictions using GNNs and other DL Models.

Prediction Model	Task	Input Data	Evaluation Criteria	Effectiveness	Limitations	Reference
Attention-Enhanced GAN (AEMP-GAN)	Flood forecasting via rainfall correction	Global Forecast System data, CPC precipitation data	RMSE, IOA, NSE, PBIAS	Improved RMSE by 28.7 %, NSE by 8.42 %	Overestimation of flow, limited peak flow correction	[37]
Bidirectional-LSTM Network	Hourly rainfall volume prediction	Weather time-series data from UK cities	Loss, RMSE, MAE, RMSLE	Comparable performance to Stacked-LSTM, RMSE: 0.0377–0.0099	Struggles with abrupt variations in precipitation patterns	[38]
Wasserstein GAN (WGAN)	Generating realistic climate states	PLASIM climate model data	Wasserstein distance, SVD, geostrophic balance	Reproduced geostrophic balance, low computational cost	Noisy geopotential fields, difficulty in fitting small-scale patterns	[39]
ConvNet and LSTM Network	Monthly rainfall prediction	Microwave, infrared, and rain gauge data	RMSE, MAPE	Similar performance to ConvNet, RMSE: 0.0360	Limited to monthly predictions, less effective for short-term forecasts	[43]
Intensified LSTM Network	Daily rainfall prediction	Weather data from Hyderabad, India	RMSE, Accuracy, Loss, Learning Rate	Best performance among compared models, RMSE: 0.0107	Overfitting, limited generalization to other regions	[44]
Variational Autoencoder (VAE)	Generating dynamics of a simple GCM	Weather state data	Reconstruction error, geostrophic balance	Reproduced geostrophic balance, low computational cost	Limited to simple GCMs, difficulty in scaling to complex models	[45]
WaveNet-LSTM, Traditional LSTM	Seasonal Precipitation Forecasting	ENSO, SOI, IOD, MJO, MEI, meteorological data	R^2 , NRMSE, FAR, CSI	WaveNet-LSTM outperformed traditional LSTM across all seasons (e.g., $R^2 = 0.91$, NRMSE = 8.68 %)	Requires high computational power	[35]
Wavelet-CNN-LSTM, Wavelet-LSTM	Climate Variability & Forecasting	Precipitation, T_{mean} , DTR across five Thai regions	R^2 , RMSE, MAPE	High accuracy in precipitation ($R^2 = 0.83$ in eastern region); consistent for T_{mean} & DTR	Struggles with precipitation forecasting in some regions	[36]
BWT-LSTM-RNN, BWT-RBFNN, MLP-ANN	Daily Precipitation Forecasting	BWT-transformed meteorological data	R^2 , MAE Pearson' Correlation RMSE, MAPE	BWT-LSTM-RNN showed highest performance ($R^2 = 0.96$ in Model M-2)	Computational complexity and overfitting	[20]

2.2. Spatio-Temporal graph neural networks (ST-GNNs)

The foundation of ST-GNNs is based on capturing spatial relationships using GNNs and sequential dynamics using recurrent/temporal layers [23,27]. This combination allows ST-GNNs to cope with the challenges of data being inherently structured as a graph and evolving [28]. GNNs were first introduced in theory by Scarselli et al. (2009), who introduced the message-passing mechanism for information aggregation from neighboring nodes on a graph [48]. Such formalization with GCNs by Kipf and Welling (2017) was then used to create convolution operations on graph-structured data based on spectral graph theory [49]. Since then, GCNs have been a fundamental component of spatial modeling in STGNNs, which allows extracting localized features of graph nodes. While GCNs can be effective on some graph-based tasks, they are unsuitable for tasks involving temporal reasoning; hence, temporal layers such as RNNs, LSTM, and GRUs are combined with GCNs [23]. Like Yu et al, ST-GCNs extend GCNs to video action recognition via learning from spatial dependencies among joints and temporal dependencies between frames. These studies emphasize that ST-GNNs are flexible in dealing with various spatio-temporal tasks [50]. Moreover, ST-GNNs have advanced further in recent years. For example, Kong et al employed GATs within the spatio-temporal framework, permitting weight-dependent weighting on related nodes. It helps the model focus more on important spatial relationships [51]. Wu et al. (2021) investigated GNNs–transformer hybrid architectures for more substantial modeling of long-range temporal dependencies [52]. ST-GNNs have been used for rainfall prediction, air quality monitoring, and climate modeling in meteorological applications [27,53]. The ability of ST-GNNs to predict precipitation patterns using spatial correlations of the weather stations and temporal trends in historical data is significant. With such studies, ST-GNNs show promise as a tool for addressing real-world questions surrounding environmental science [53].

In this study, an ST-GNN is developed to predict rainfall at weather stations in Thailand (Fig. 3). Spatial relationships are represented using graph-based operations, and temporal dynamics are processed by RNNs embedded in the model (Fig. 4). The meteorological features include

maximum temperature, minimum temperature, relative humidity, wind speed, wind direction, and precipitation from multiple weather stations. The preprocessing step is to create a spatial graph where weather stations are nodes and adjacency between them is defined by a k nearest neighbors ($k = 5$) adjacency matrix. Z-score normalization normalizes the input features to zero mean and unit variance [54]. The mean precipitation value on all dates per station is the target variable. The model architecture has three main components: spatial processing, temporal processing, and fully connected layers, as presented in Table 2. Two GCN layers and a GAT layer are used to achieve spatial processing. GCN layers use input data-passing mechanisms to aggregate information from neighboring nodes, and residual connections preserve information from earlier layers [55]. The GAT layer aggregates the spatial information (adapting to their neighbors) by assigning attention weights for neighboring nodes [56]. The bidirectional GRU handles temporal processing and captures temporal dependencies in data [51]. Secondly, the learned representations are mapped to the final output, corresponding to the predicted rainfall values with two fully connected layers. Adam optimizer with an initial learning rate of 0.01 is used, and the gradients are clipped to prevent exploding gradients during training [57]. The learning rate scheduler reduces the learning rate when the loss plateaus to stabilize the convergence. By modeling spatial and temporal dependencies in meteorological data, the proposed ST-GNNs are effective. The model integrates GCN, GAT, and GRU layers, achieving accurate rainfall prediction. It also provides interpretable insights from the relationships between weather stations.

2.3. Generative adversarial graph neural networks (GA-GNNs)

GA-GNN is inspired by the adversarial training framework used in this model, which is based on GANs and GNNs [58]. GA-GNN is based on GNNs, generalizing neural networks for graph-structured data. GCNs and GATs have captured spatial dependencies in different domains, including traffic forecasting [41] and molecular property prediction [59, 60]. A GAN consists of a generator and a discriminator, where the generator tries to learn to produce realistic data samples, and the discriminator tries to differentiate them from actual data samples [58,

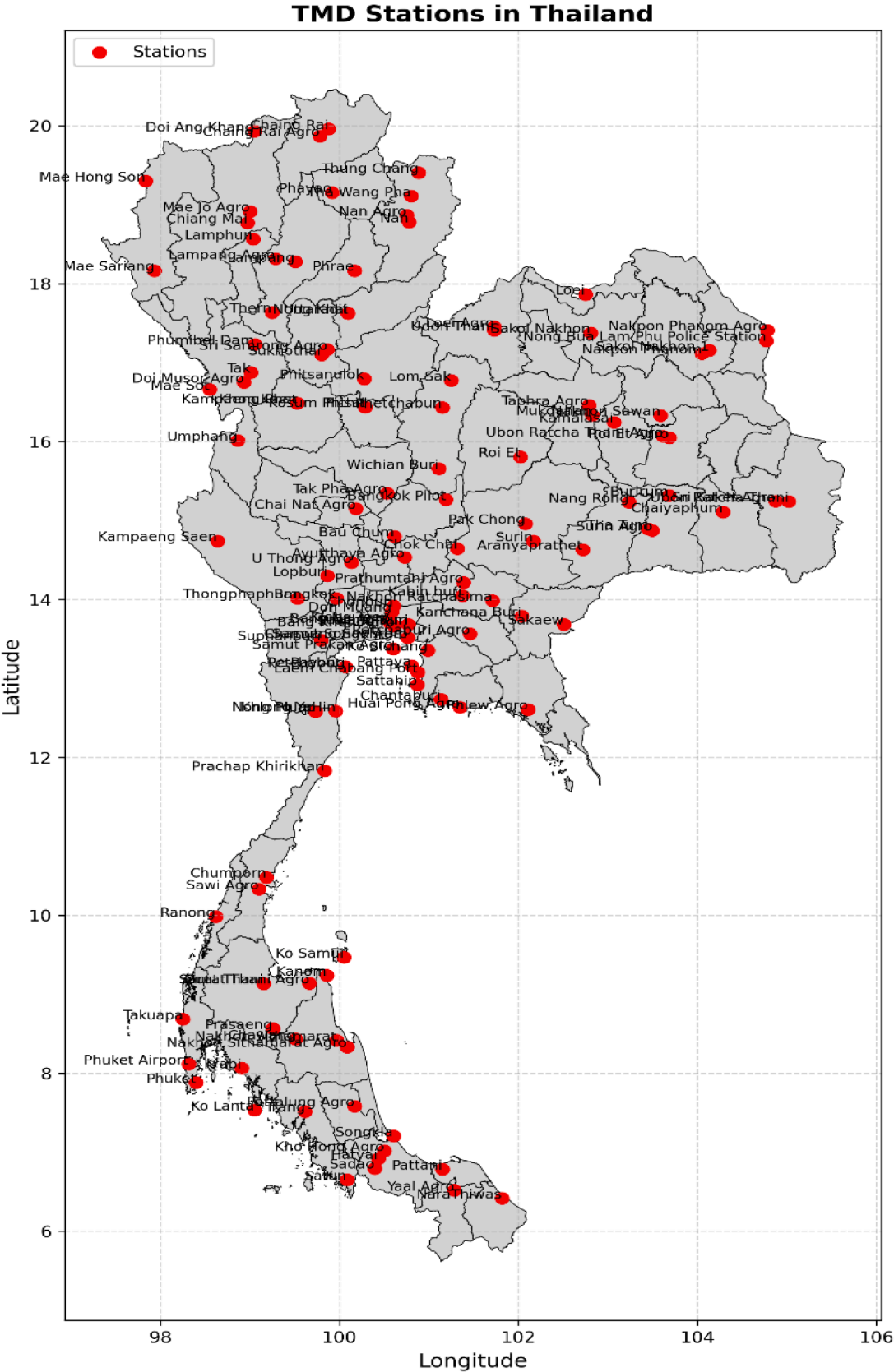
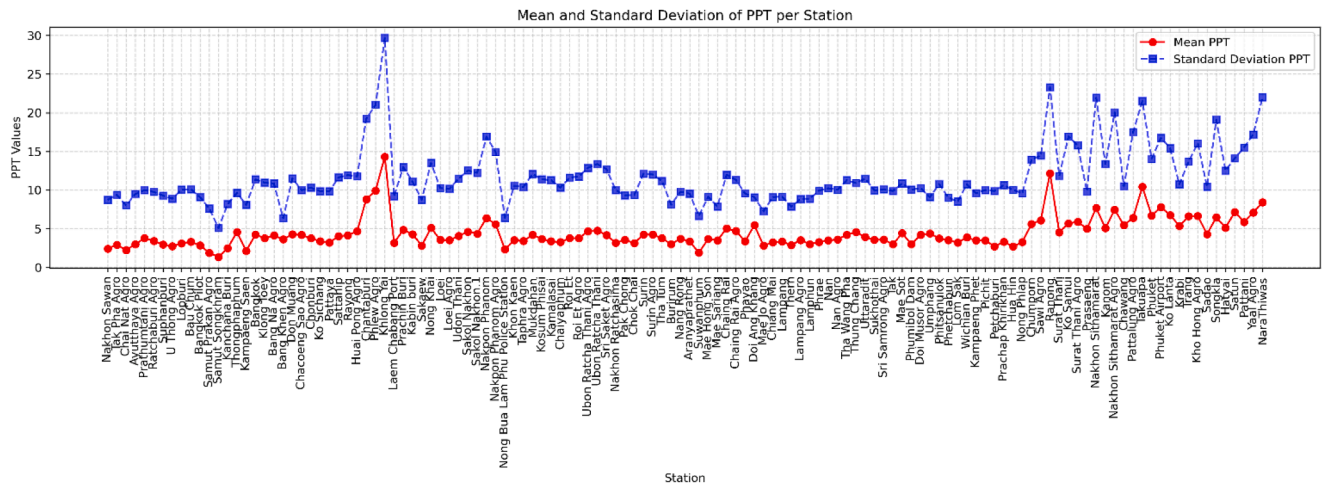


Fig. 1. Distribution of TMD observation stations over Thailand.



(a) mean and standard deviation of precipitation (PPT) per TMD station

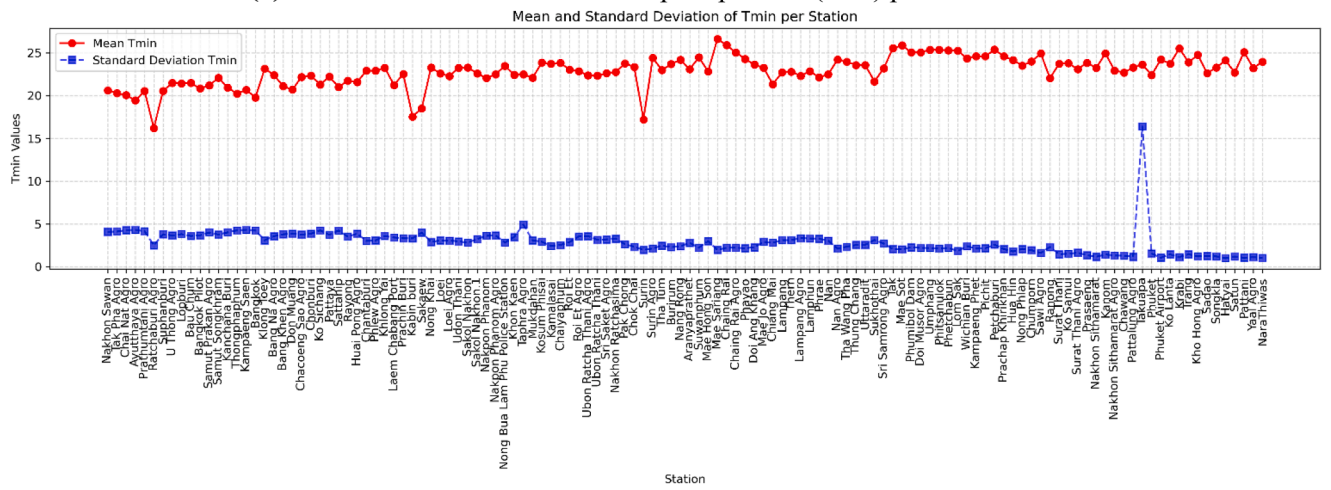
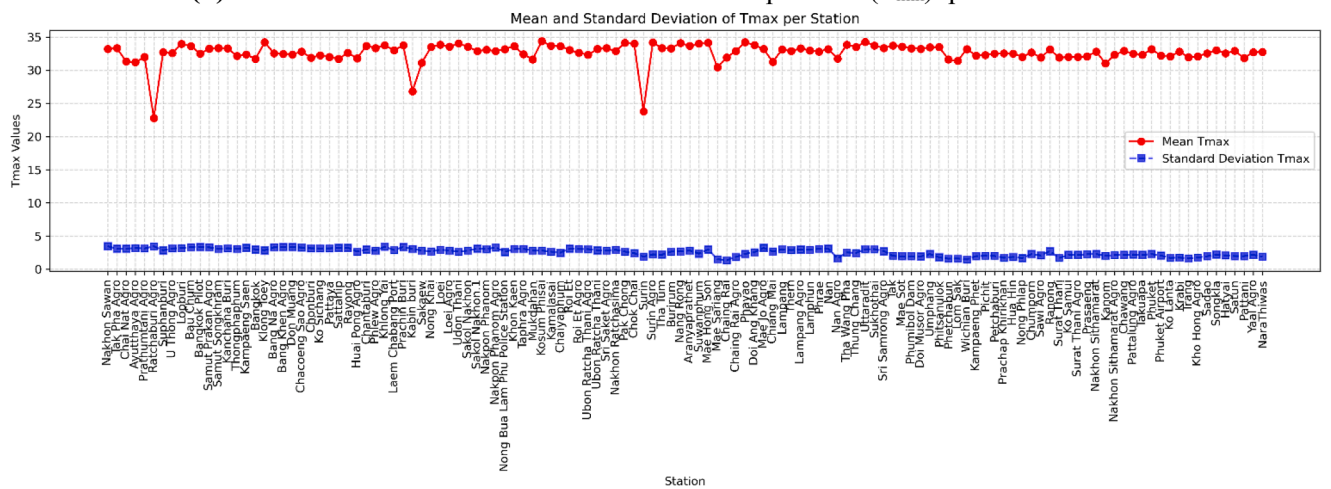
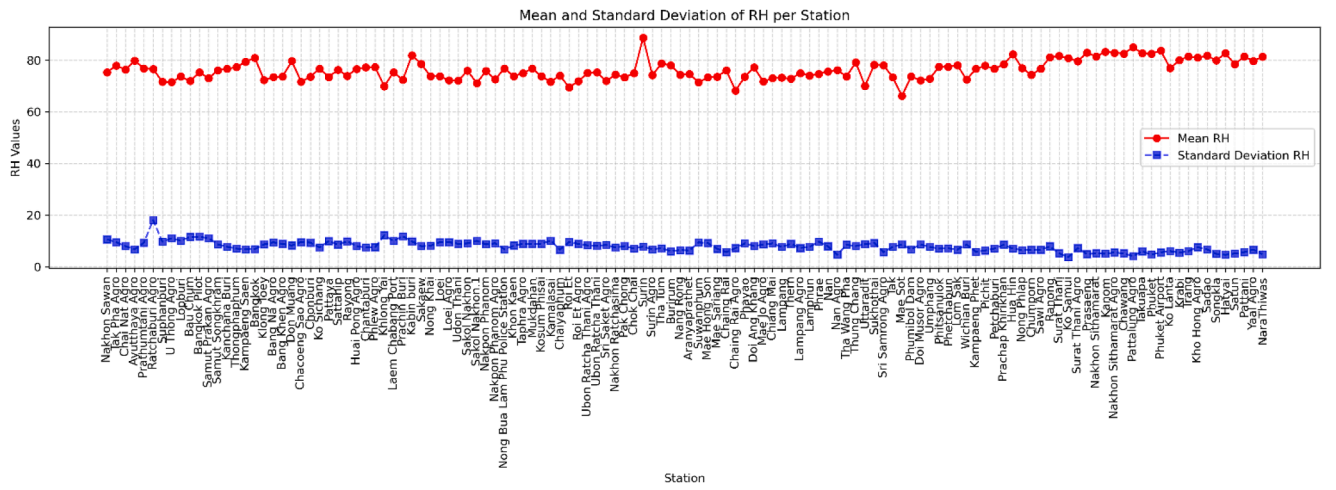
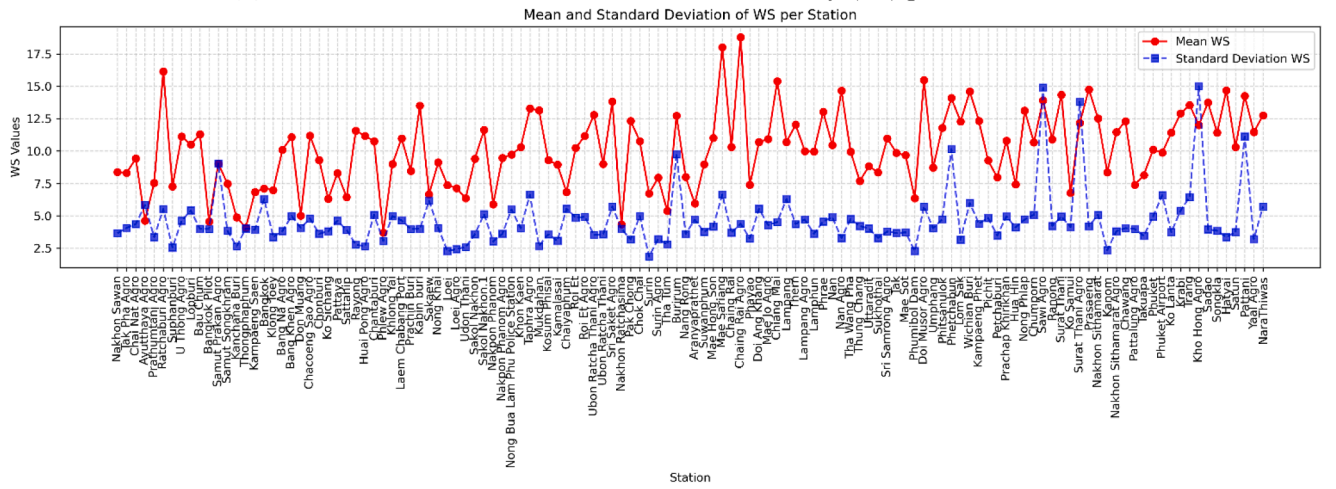
(b) mean and standard deviation of minimum temperature ($T_{\min}$) per TMD station(c) mean and standard deviation of maximum temperature ($T_{\max}$) per TMD station

Fig. 2. Mean and SD of all selected variables (a-f) from 1993–2022.

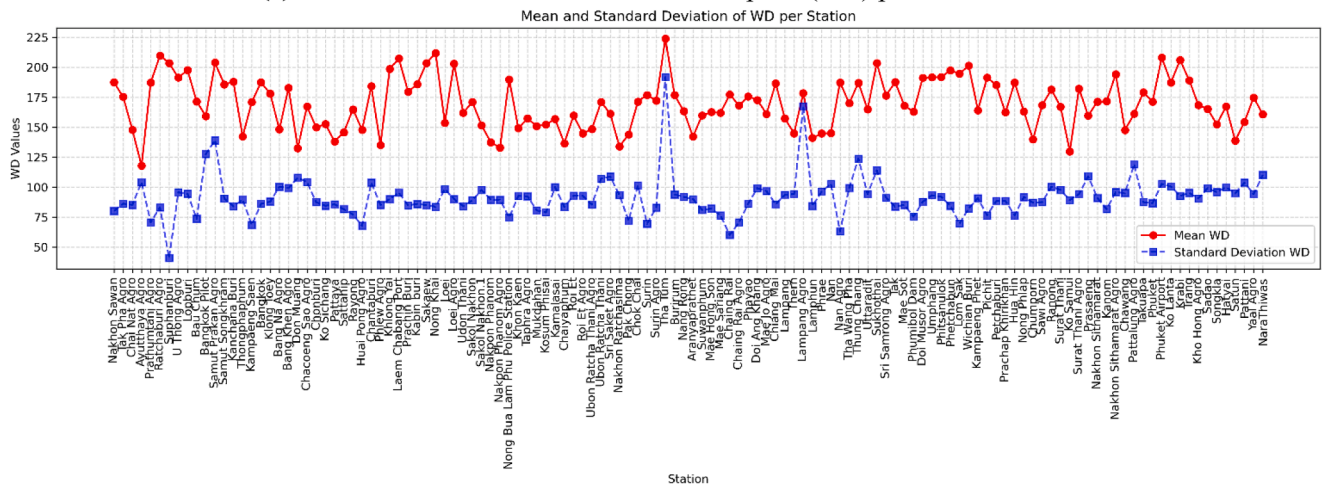
61]. In regression tasks, like weather prediction, this adversarial interplay has been adapted to generate more realistic and accurate outputs [34]. The GA-GNN model addresses the limitations of conventional methods that are deficient in capturing the complex correlations and temporal dynamics intrinsic to meteorological data by combining GNNs

with GANs [28,62,63]. Many researchers have explored GA-GNN for rainfall prediction with more precise outcomes [34,40,61].

In this study, GA-GNNs use the spectral graph theory to propagate input data through nodes and introduce the attention mechanism that allows the model to dynamically pay attention to the critical spatial

(d) mean and standard deviation of relative humidity (R_H) per TMD station

(e) mean and standard deviation of wind speed (WS) per TMD station



(f) mean and standard deviation of wind direction (WD) per TMD station

Fig. 2. (continued).

relationships [56]. The preprocessing step starts by processing meteorological input temporal and spatial datasets from different stations, including PPT, $T_{\max}$, $T_{\min}$, R_H , WS, and WD. Then, these features are reshaped into a long format and standardized to be uniform. An adjacency matrix derived from the geographical coordinates of stations represents spatial relationships between stations and is then transformed

into a graph structure (See Fig. 5). The meteorological stations are represented as graph nodes, and the spatial proximity is encoded as an edge, making the model effective in learning spatial correlations. GCN and GAT layers are used in the generator part of the GA-GNN to process node features and edge indices, respectively, to learn local and global dependencies in the graph. The discriminator checks whether generated

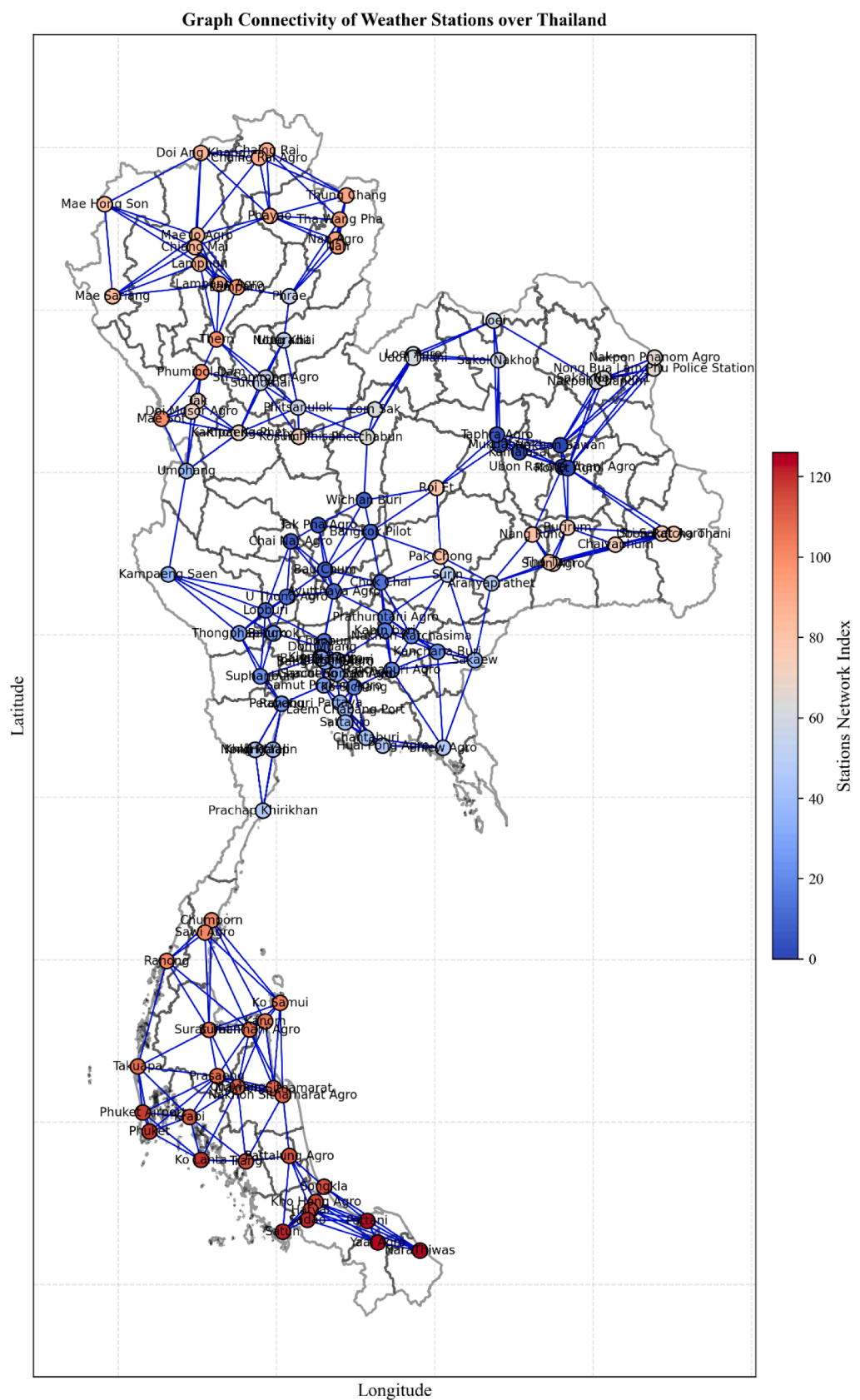


Fig. 3. Spatial relationships are represented using graph-based operations.

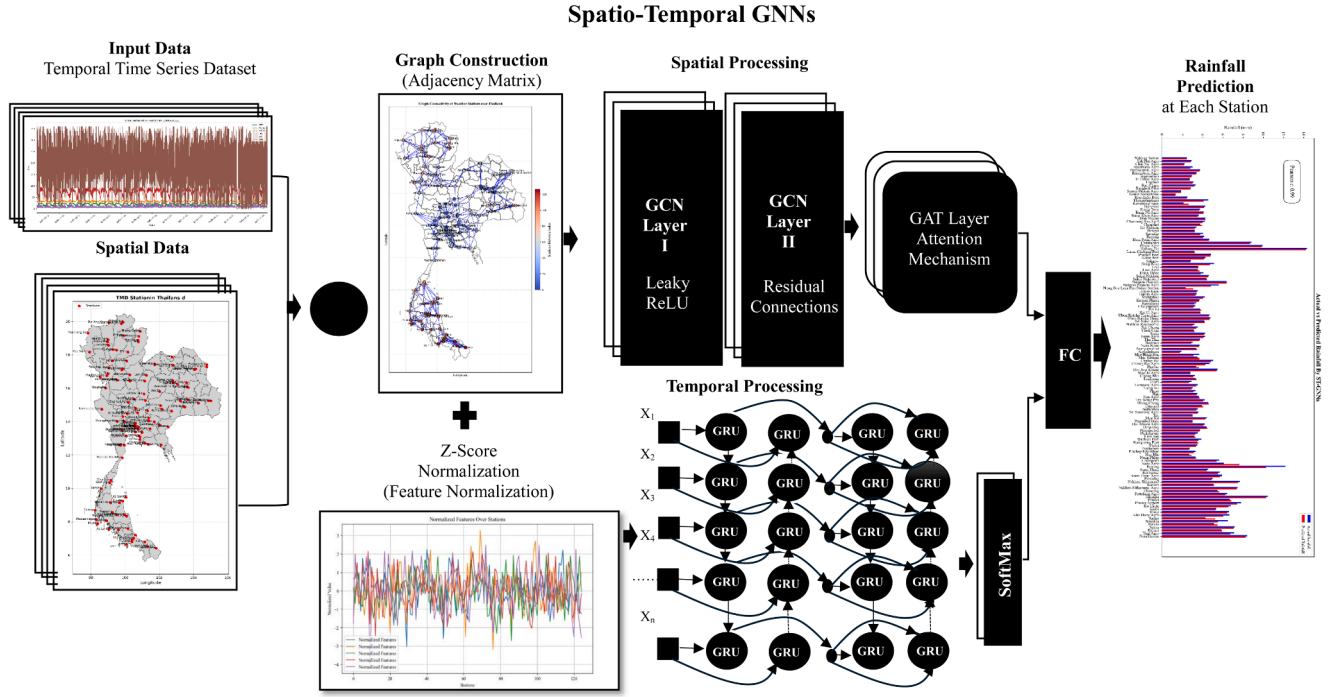


Fig. 4. Working mechanism of ST-GNNs for rainfall prediction.

Table 2
Architecture of ST-GNNs developed for rainfall prediction.

Layer	Operation/Mathematical Expression	Output Shape
Input Layer	Node Features ($X \in \mathbb{R}^{N \times F}$) (1)	$N \times F$
GCN Layer 1	$h_i^{(l+1)} = \sigma \left(\sum_{j \in N(i)} \frac{1}{\sqrt{\deg(i)} \times \deg(j)} \times W h_j^{(l)} \right)$ (2)	$N \times H$
BatchNorm + LeakyReLU	Batch normalization followed by LeakyReLU activations	$N \times H$
GCN Layer 1	Residual Connection = $h_i^{(l+1)} = \sigma \left(h_i^{(l)} \times GCN(h_i^{(l)}) \right)$ (3)	$N \times H$
GAT Layer	Attention Mechanism = $h_i^{(l+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij} W h_j^{(l)} \right)$ (4)	$N \times K \times K$
GRU Layer	Temporal Processing = $h_t = (1 - z_t) \times h_{t-1} + z_t \times \tilde{h}_t$ (5)	$N \times H$
Fully Connected Layers	Linear Transformation = $y = W_2(LeakyReLU(W_1 \cdot x))$ (6)	$N \times 1$

rainfall predictions are realistic; the model generates outputs like actual rainfall patterns. The overall working mechanism of the GA-GNNs is provided in Fig. 6. Mathematically, this model can be expressed as:

$$GCN \text{ Layer } = H(l+1) = \sigma \left(\tilde{D} - \frac{1}{2} \tilde{A} \tilde{D} - \frac{1}{2} h^l * w^l \right) \quad (7)$$

Where, $\tilde{A}$ is the adjacency matrix with added self-loops, $\tilde{D}$ is the degree matrix, h^l represents the feature matrix at layer l , and w^l denotes the weight matrix.

$$GAT \text{ Layer Attention Coefficient } = e_{ij} = LeakyReLU(a^T [W h_i || W h_j]) \quad (8)$$

where a is the attention mechanism, W is the weight matrix, and h_i and h_j are feature vectors of nodes i and j .

$$Generator \text{ Loss Function } = L_G = L_{GAN} + \lambda L_{L1} \quad (9)$$

Where λ controls the trade-off between the two components.

$$Discriminator \text{ Loss Function } = L_D = L_{real} + L_{fake} \quad (10)$$

where L_{real} and L_{fake} correspond to real and fake data, respectively. Furthermore, the detailed hyperparameters comparison is provided in Table 3.

3. Results and discussion

This section presents the comprehensive results of ST-GNNs and GA-GNNs for rainfall prediction across Thailand. This study employed evaluation metrics to assess model performance, including the R^2 , MSE, MAE, RMSE, RRMSE, NRMSE, and MAPE. SHAP analysis was used to quantify input variables' contributions to the prediction's accuracy. Then, rainfall predictions for each meteorological station were evaluated, and the prediction model was thoroughly compared with relevant benchmark studies.

SHAP analysis results for a combination of ST-GNNs give insights into the comparative importance and the effect of various input variables on the predicted model output (Fig. 7). The analysis shows that relative humidity (R_H) has the highest positive impact on rainfall prediction, with a SHAP value of 0.634, meaning that higher R_H significantly impacts increasing rainfall. R_H is important in cloud formation and precipitation processes and is consistent with meteorological principles. A SHAP value of 0.465 also indicates that WS positively impacts atmospheric dynamics that are favorable for rainfall.

The SHAP value of T_{max} is moderately positive (0.510) because it increases atmospheric moisture holding capacity and convective activity. On the other hand, T_{min} has a negative SHAP value (−0.364), indicating that lower nighttime temperatures may reduce rainfall by stabilizing the atmosphere. The SHAP value of WD is 0.183, which shows that it has a more negligible positive impact (modulating rainfall patterns) and, hence, a secondary role. These findings are consistent with the complexity of meteorological variables in the tropics, where R_H and WS are key drivers of precipitation. SHAP analysis reveals that the model can convey these relationships, making it a viable means of unpacking the contribution of variables in rainfall prediction.

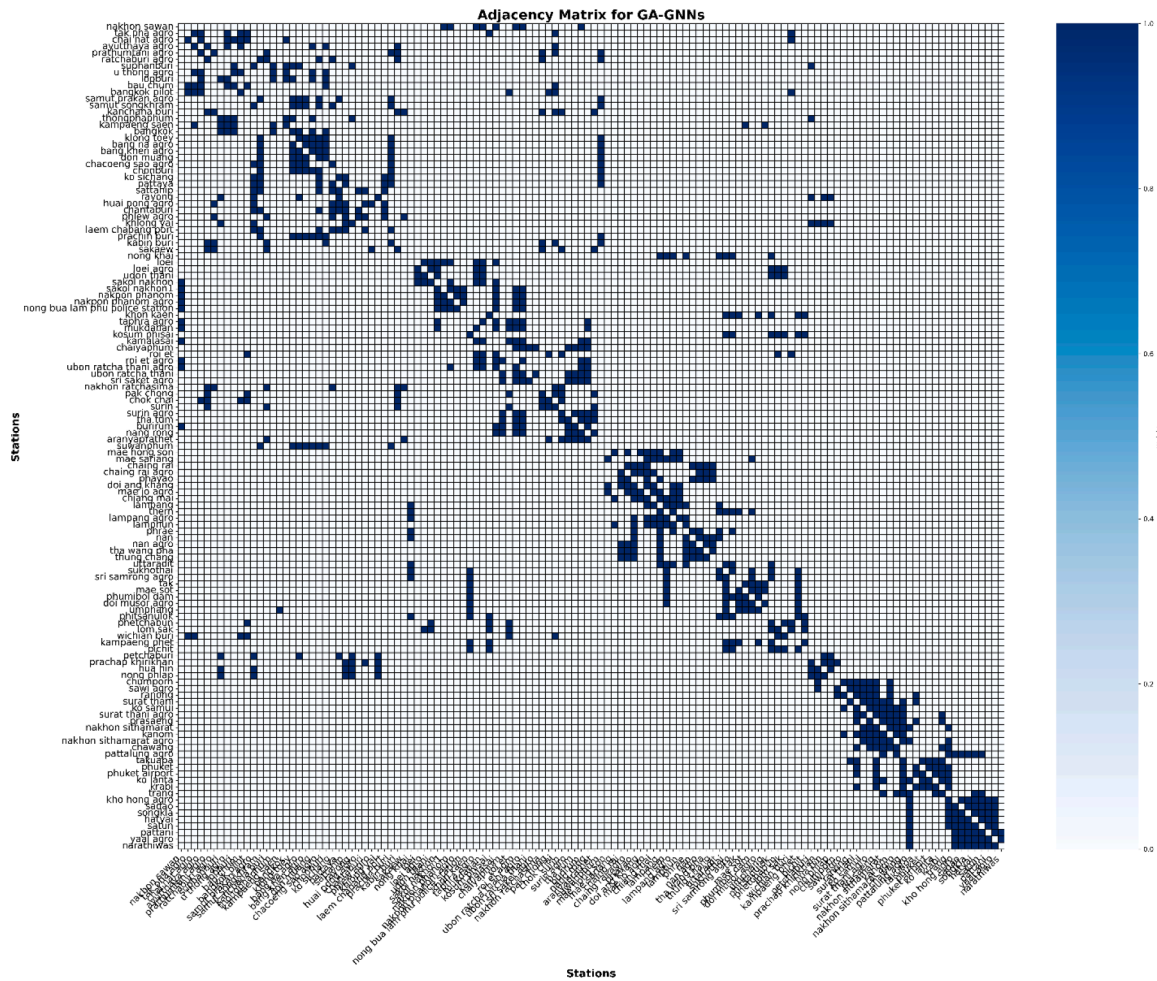


Fig. 5. Adjacency matrix for the GA-GNNs model derived from the geographical coordinates of stations represents spatial relationships between stations, which transformed into a graph structure.

SHAP analysis for GA-GNNs in meteorological stations is shown in Fig. 8, which shows how different input variables contribute to the model's output. The results show that WD, R_H , and WS are the most important factors, as their high SHAP values indicate that they significantly affect rainfall prediction. T_{max} also has a significant positive effect in improving convective activity and atmospheric instability. On the other hand, T_{min} has a negative impact, which implies that lower nighttime temperatures can stabilize the atmosphere and decrease the probability of rainfall. The influence of WS and interactions like R_H -WS and T_{max} - T_{min} vary, suggesting the complicated meteorological variables in tropical regions. SHAP values of -0.075 to 0.125 indicate that the model can capture these complex relationships and offer a solid basis for interpreting variable contributions in rainfall prediction with GA-GNNs.

Both ST-GNNs and GA-GNNs models are evaluated in terms of performance metrics, as shown in Fig. 9. The R^2 for ST-GNNs is 0.9580, which shows good agreement between predicted and observed rainfall and the high predictive accuracy of the model. The MAE of 0.3950 and the RMSE of 0.1560 are good indicators of the model's precision since they have low error, implying slight variation from actual rainfall measurements. The model is robust as the MAPE of 3.5410 % and NRMSE of 6.3980 % also indicate minor percentage errors and normalized deviations.

On the other hand, the GA-GNNs have an R^2 value of 0.5800 and are farther away from the diagonal, suggesting a moderate correlation between its predictions and observations. At 1.5770 and 0.8400, GA-GNNs have higher MAE and RMSE values, respectively, showing more prediction errors than ST-GNNs. Moreover, GA-GNNs have higher MAPE

and NRMSE of 3.4730 % and 6.3980 %, respectively, which further supports the reduced accuracy of this model. It is confirmed by the higher MSE of 2.4800 for GA-GNNs compared to 0.0200 for ST-GNNs in minimizing prediction errors.

These findings indicate that ST-GNNs better represent rainfall pattern spatio-temporal dynamics in Thailand since they can accurately deal with temporal dependencies and spatial relationships. Despite giving valuable insights, GA-GNNs do not effectively capture the interplay of geography, time, and reaching, as they may suffer from more significant prediction errors. ST-GNNs' superior performance is consistent with their design, designed to handle both spatial and temporal data, which is more appropriate for rainfall prediction tasks in regions with complex meteorological conditions, such as Thailand.

ST-GNNs are used to predict the rainfall at Thailand meteorological stations, and Figs. 10(a) and (b) present a detailed comparison of actual versus predicted rainfall with station-level performance. Fig. 10(b) indicates a strong correlation between actual and predicted rainfall (r) 0.99. This high correlation suggests that ST-GNNs are very good at capturing rainfall's spatial and temporal dynamics and making predictions close to observed data at multiple stations. For example, the model can accurately predict the northern and northeastern Thailand stations, which are subject to large seasonal rainfall variability, demonstrating the model's capability to deal with complex climatic patterns. The spatial distribution of rainfall predicted by ST-GNNs across different latitudes and longitudes is further corroborated in Fig. 10(a), a map of predicted rainfall based on ST-GNNs. It implies that ST-GNNs are good at modeling the complex interactions between geographical and

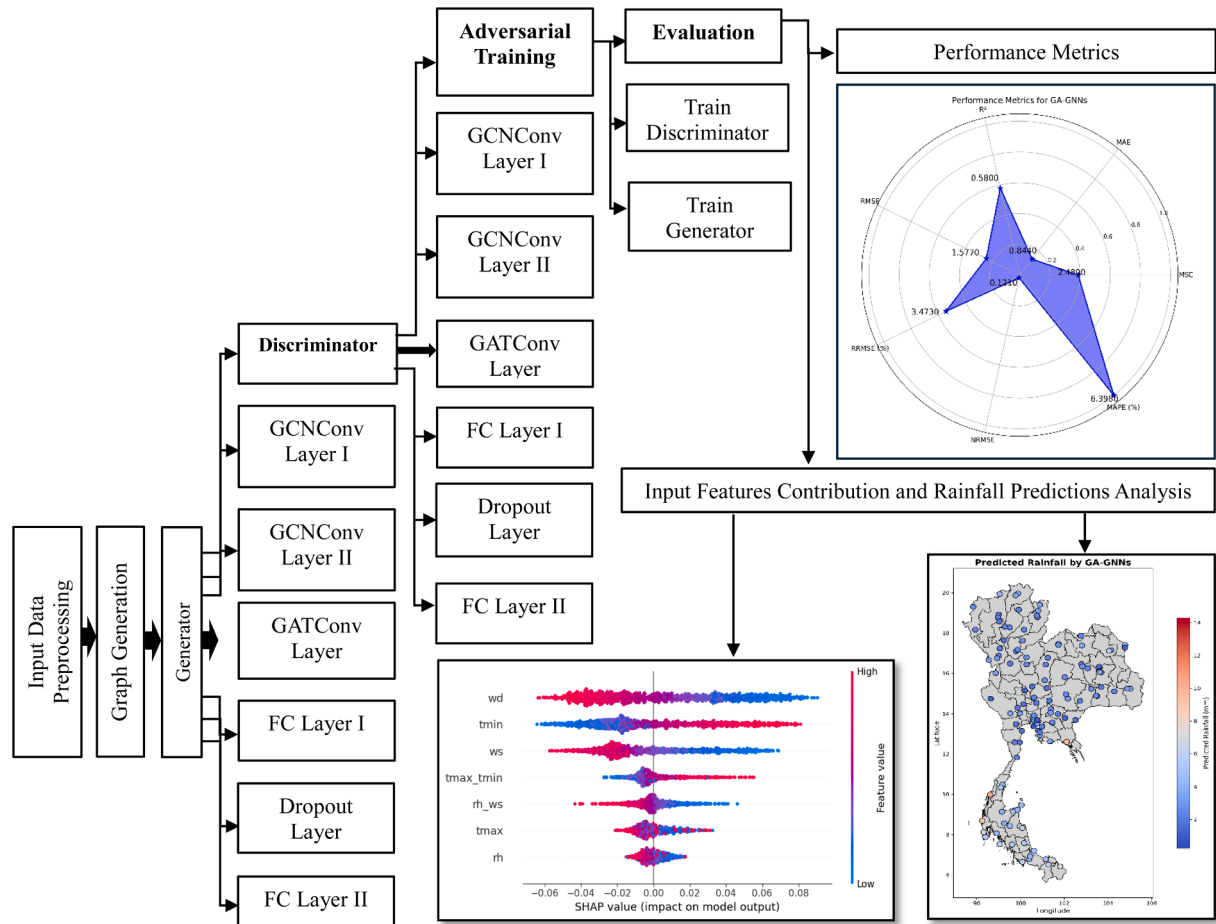


Fig. 6. Working Architecture of GA-GNNs.

Table 3
Comparative hyperparameter settings for ST-GNNs and GA-GANs.

	Hyperparameter	ST-GNNs	GA-GNNs
Graph Construction	n_neighbors	8 (for k-NN graph)	5 (for k-NN graph)
	mode	—	'connectivity' mode
	include_self	Implicit (residual connections in GCN layers)	False (no self-loops in graph construction)
	neighbor_threshold	8	—
Input Features	input_dim	Dynamic (Tmax, Tmin, RH, WS, WD, Rainfall, etc.)	Dynamic (5–7 meteorological features)
Spatial Modeling	GCN Layers	Two GCN layers with residual connections	conv1 and conv2: GCNConv(...)
	GAT Layer	One GAT layer	gat: GATConv(...)
	num_heads (GAT)	4	4
	hidden_dim	256 (for both GCN and GAT layers)	64
	dropout	0.3	0.5
	leaky_relu slope	0.01	0.2
	batch_norm	Implicit in architecture	True (applied after each GCN layer)
	contact	—	True (attention head concatenation)
Temporal Modeling	Temporal Layer	Bidirectional GRU (hidden size = $256 \times \text{num_heads}$)	Bidirectional GRU (hidden size = $64 \times \text{num_heads}$)
Fully Connected Layer	Output Layer	Two-layer MLP for final rainfall prediction	Two-layer MLP in generator output
Training Parameters	num_epochs	500	500
	optimizer_G	Adam (lr=0.001, betas=(0.5, 0.999))	Adam (default)
	optimizer_D	Adam (lr=0.0001, betas=(0.5, 0.999))	—
	learning_rate	Adaptive with StepLR scheduler	Initial: 0.01
	scheduler	StepLR (step_size=50, gamma=0.5)	ReduceLROnPlateau (factor=0.5, patience=10)
	criterion_GAN	BCEWithLogitsLoss() for adversarial loss	Included in L _D and L _G
	criterion_L1	L1Loss()	Used in combination with GAN loss
	L1_loss_weight (λ)	10 (to balance L _{GAN} and L1)	10 (as λ in $L_G = L_{GAN} + \lambda * L1$)
	criterion	Implicit through loss composition	MSELoss
	grad_clip	1.0	1.0
	early_stopping_patience	20	20
Feature Engineering	normalization	StandardScaler (zero mean, unit variance)	StandardScaler (zero mean, unit variance)
Graph Nodes	feature_engineering	Raw meteorological inputs used	Includes interaction terms: tmax_tmin, rh_ws
	num_nodes	Based on the number of weather stations	125 (fixed weather stations)

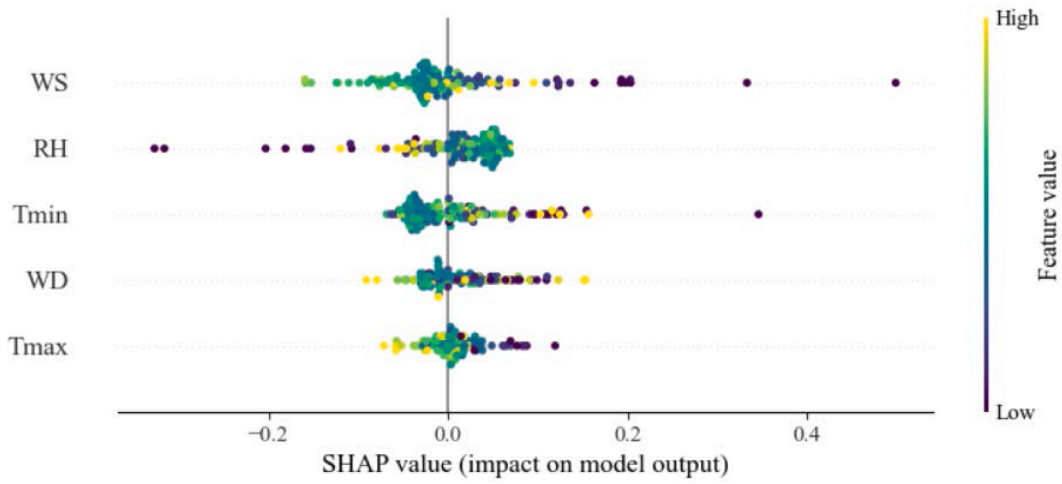


Fig. 7. SHAP analysis for ST-GNNs.

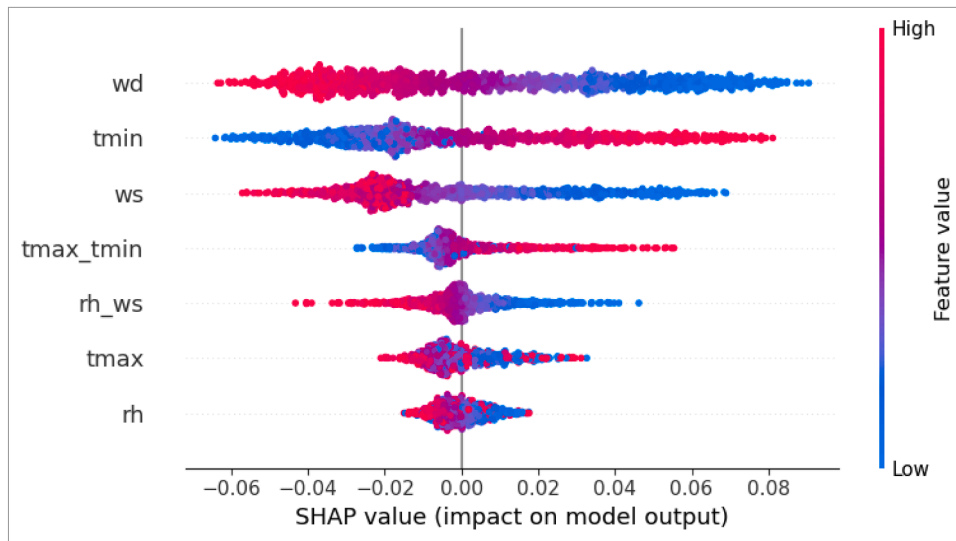


Fig. 8. SHAP analysis for GA-GNNs.

meteorological variables and, thus, are highly reliable in predicting rainfall in different climatic conditions.

On the other hand, Fig. 11(b) for GA-GNNs shows a moderate correlation between actual and predicted rainfall with Pearson r of 0.58. The lower correlation implies that GA-GNNs can partially capture some rainfall patterns, but their predictions are less accurate than ST-GNNs, especially at some stations. For example, stations on the coast of southern Thailand are affected by monsoonal winds and sea breezes, so these discrepancies are more notable between the predicted and actual numbers. Fig. 11(a), a map of the predicted rainfall by GA-GNNs, has a spatial distribution broadly consistent with actual rainfall but less precise and variable. However, due to the intricate spatiotemporal relationships that impact rainfall, GA-GNNs may fall short in accounting for all these factors, which results in even more considerable differences between predicted and actual rainfall in highly complex environments where the microclimate plays an important role.

The architecture of ST-GNNs and GA-GNNs explains the differences in their performance. This paper presents ST-GNNs designed to model both space and time dependencies better to catch the dynamic nature of rainfall over multiple stations. However, GA-GNNs may not fully integrate temporal dynamics to incorporate geographical information, leading to less accurate predictions, especially in stations with high

climatic variability. The results show that rainfall prediction models should consider spatial and temporal factors. ST-GNNs are well suited for meteorology and climate science applications, particularly for station levels. Overall, this station-level analysis highlights the significance of modeling at different scale levels for capturing localized climatic variability to produce better, more actionable rainfall precipitation estimates.

3.1. Model's evaluation for TMD's rainfall classification

Figs. 12 and 13 highlight the classification performance of ST-GNNs and GA-GNNs for rainfall prediction across Thailand. The models' performance differs between the TMD rainfall categories used for the classification. It implies that ST-GNNs can achieve a classification accuracy of 0.992, much higher than the 0.611 GA-GNNs. The F1 scores of ST-GNNs are near perfect: 0.995 for light rain, 1.000 for moderate rain, and 0.933 for no rain or drizzle. These results suggest that the model forecasts much more of the rainfall patterns than it misclassifies.

On the other hand, GA-GNNs are not good, achieving an F1 score of 0.729 for light rain, 0.667 for moderate rain, and 0.237 for no rain or drizzle. GA-GNN performs poorly distinguishing light and moderate rainfall, with recall values of 0.500 and 0.595 for moderate and light

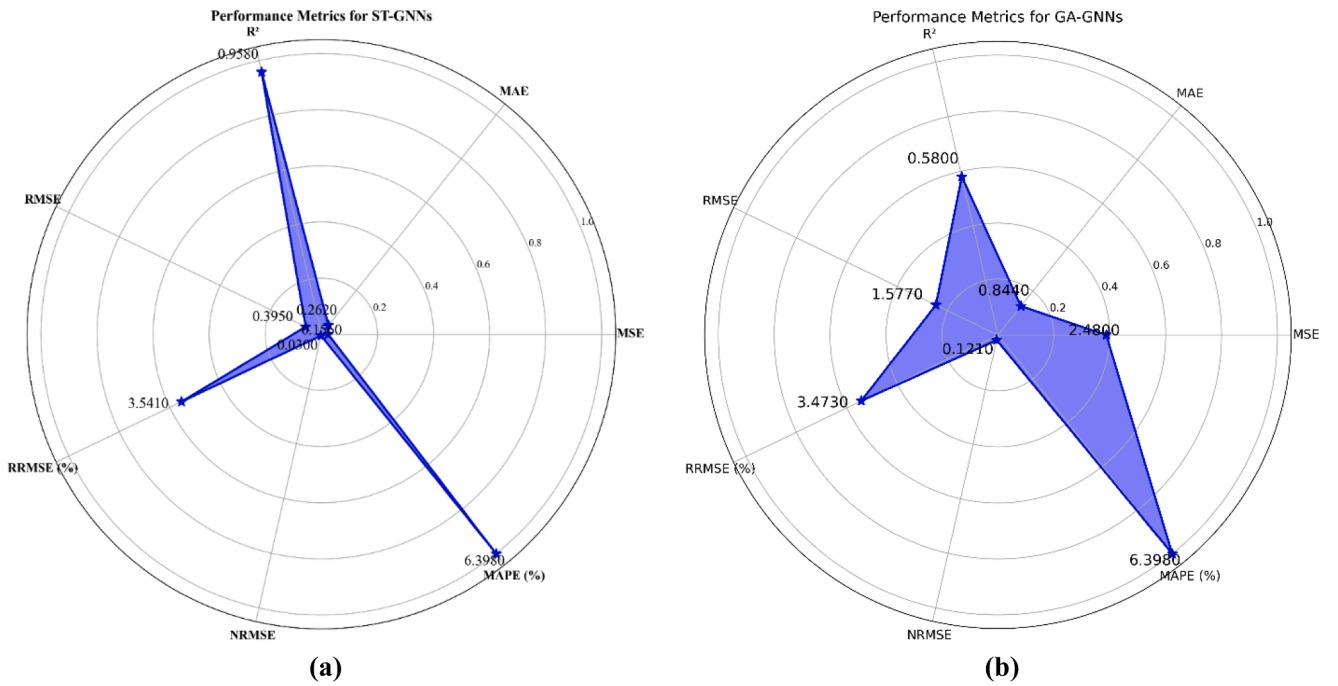


Fig. 9. (a) Evaluation results for ST-GNNs (b) GA-GNNs.

rainfall, respectively. Frequent underestimation or overestimation of rainfall intensities follow. It also highlights how ST-GNNs capture the spatiotemporal dependencies of rainfall more effectively so that predictions can provide more reliable classifications. However, GA-GNNs exhibit a high tendency for misclassification, especially when distinguishing dry and wet conditions. In addition, the precision for no rain or drizzle is exceptionally low in GA-GNNs (0.135) compared to 0.875 in ST-GNNs, which further demonstrates the superiority of ST-GNNs. The results indicate that ST-GNNs are more accurate and reliable for the rainfall classification in Thailand and, therefore, more suitable for meteorological applications. Due to their capabilities to represent complex spatial and temporal relationships, higher classification accuracy is achieved, and thus, it helps in better weather forecasting and assessing related climate risk in diverse climatic regions.

3.2. Discussion

The study's results show that ST-GNNs outperform GA-GNNs in terms of predictive accuracy with R^2 of 0.9580 against 0.5800 for GA GNNs. The results of this study are consistent with previous studies that found spatio-temporal models effective in capturing complex meteorological patterns. For example, Wu et al. (2022) employed inductive ST-GCNs for short-term quantitative precipitation forecasting and achieved high accuracy by modeling spatial relations in Eastern China rain gauges [23]. Similarly, Yang et al. (2022) also showed how spatio-temporal graph-guided convolutional LSTM networks can be effectively used for tropical cyclone precipitation nowcasting and highlighted the benefits of the model and recommended use in the rainfall prediction model [27]. The results of both studies demonstrated that ST-GCNs and ST-GNNs outperformed other traditional GNNs as they can model both temporal and spatial dynamics.

On the other hand, although GA-GNNs are novel, they had lower predictive accuracy, as indicated by higher MAE (1.5770) and RMSE (0.8400) than ST-GNNs. This inconsistency could result from the difficulties faced by adversarial training in representing the complex and relative spatio-temporal relations needed for proper rainfall prediction. In another experimental study by Luo et al. (2022) on precipitation nowcasting using generative adversarial networks, they also described

limitations where the model often failed to produce small-scale precipitation patterns correctly [34]. In this study's findings, the results of GA-GNNs on generating realizable rainfall scenarios are promising. However, these algorithms need to be trained to achieve the accuracy of this study's ST-GNNs under more complex atmospheric conditions.

Additionally, the SHAP analysis allows for understanding the relative importance of the different meteorological variables in rainfall prediction. The most influential variable in the case of ST-GNNs was WS, with a SHAP value of 0.634, followed by R_H and T_{min} . The moderate positive influence of T_{max} on convective activity and atmospheric instability can account for the moderate positive influence of T_{max} on the averaged and ensemble mean precipitation, as in previous studies such as Waqas et al. (2024) highlighted the need for these variables in their DL framework for precipitation forecasting in Thailand, as it is important to consider several meteorological factors to obtain accurate predictions [35].

Through the SHAP analysis for GA-GNNs, we also identified R_H and WS as significant predictors; however, the model had an inadequate ability to learn the high dimensional dependencies between variables compared to ST-GNNs. These outcomes are consistent with the work of Venkatesh et al. (2021), who observed that GANs generate realistic rainfall patterns but cannot reproduce the fine spatial and temporal dependencies needed for concretely predicting rain [64]. This observation is further corroborated by the work of Kim and Hong (2021), where their conditional generative adversarial network for very short-term rainfall prediction presented promising results but had difficulty generalizing over different climatic conditions [40]. The comparative analysis of ST-GNNs and GA-GNNs was conducted in the areas where the spatial and temporal dynamics need to be integrated into the rainfall prediction models. This is because ST-GNNs outperform the traditional approaches in predicting rainfall based on accurately computing spatial relationships and temporal evolution, which are crucial in areas with high climatic complexities, such as Thailand. As Peng et al. (2020) also demonstrated, spatial-temporal incidence dynamic GNNs have the advantage of handling complex data structures and are well-suited for spatial-temporal models [26].

While GA-GNNs have difficulty integrating temporal dynamics to obtain predictive precision. The drawback that Besombes et al. (2022) reported is that GANs can reproduce geostrophic balance but fail at the

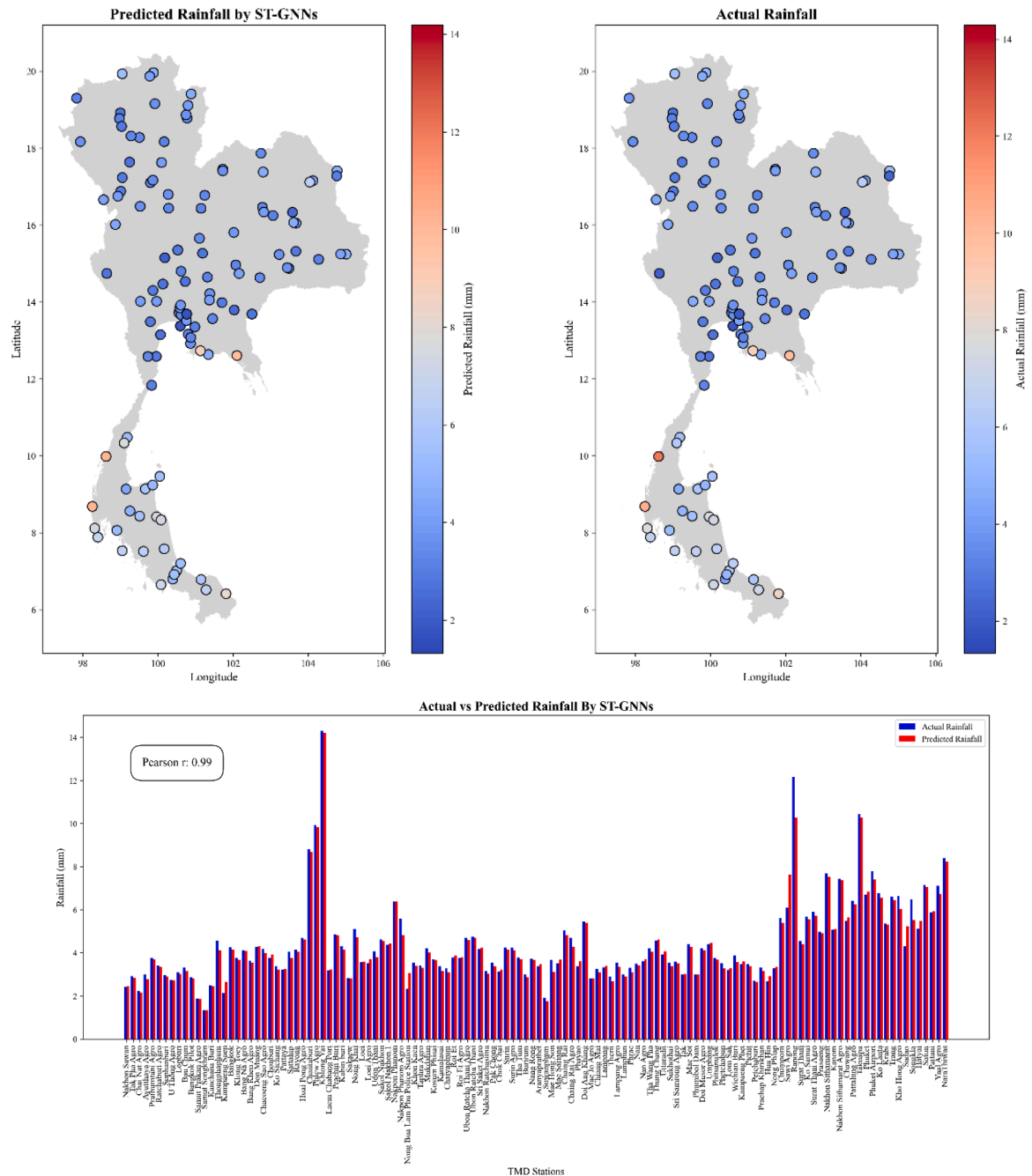


Fig. 10. Actual vs predicted rainfall by ST-GNNs (a) geospatial comparison, and (b) actual vs. predicted rainfall for each meteorological station with Pearson correlation.

resolution of small scales [39]. Luo et al. (2022) also showed that using GANs for precipitation nowcasting is challenging due to the need to capture fine spatial and temporal dependencies for accurate predictions [34]. These results suggest that GA-GNNs are a promising approach to generating realistic rainfall scenarios, but further polishing is needed to achieve results that are as good as ST-GNNs. The results of this study demonstrate that ST-GNNs are useful for rainfall prediction in complicated climatic conditions in Thailand. GA-GNN is expected to produce accurate rainfall scenarios, but it needs to be refined to match the level of accuracy of ST-GNNs. The results of this study include improvements to disaster preparedness and water resource management, and they suggest the need for robust models that can represent the complex spatio-temporal spatial pattern of rainfall. Future work should enhance the robustness of GA-GNNs and hybrids by combining the best ST-GNNs

and GA-GNNs for better rainfall prediction.

4. Conclusion

Accurate rainfall forecasting has gained increased significance amidst the current climate change and global warming that has increased the occurrence and intensity of severe rainstorms, thus increasing the threat to water resource management and disaster mitigation. This paper thus examines two deep modeling systems, ST-GNNs and GA-GNNs, to forecast rainfall using a complete Thai meteorological data set of 1993–2022.

The empirical findings reveal that the ST-GNN model is better than the GA-GNN alternative in terms of forecasting accuracy: the former model showed an R^2 of 0.9580, which is indicative of the strong

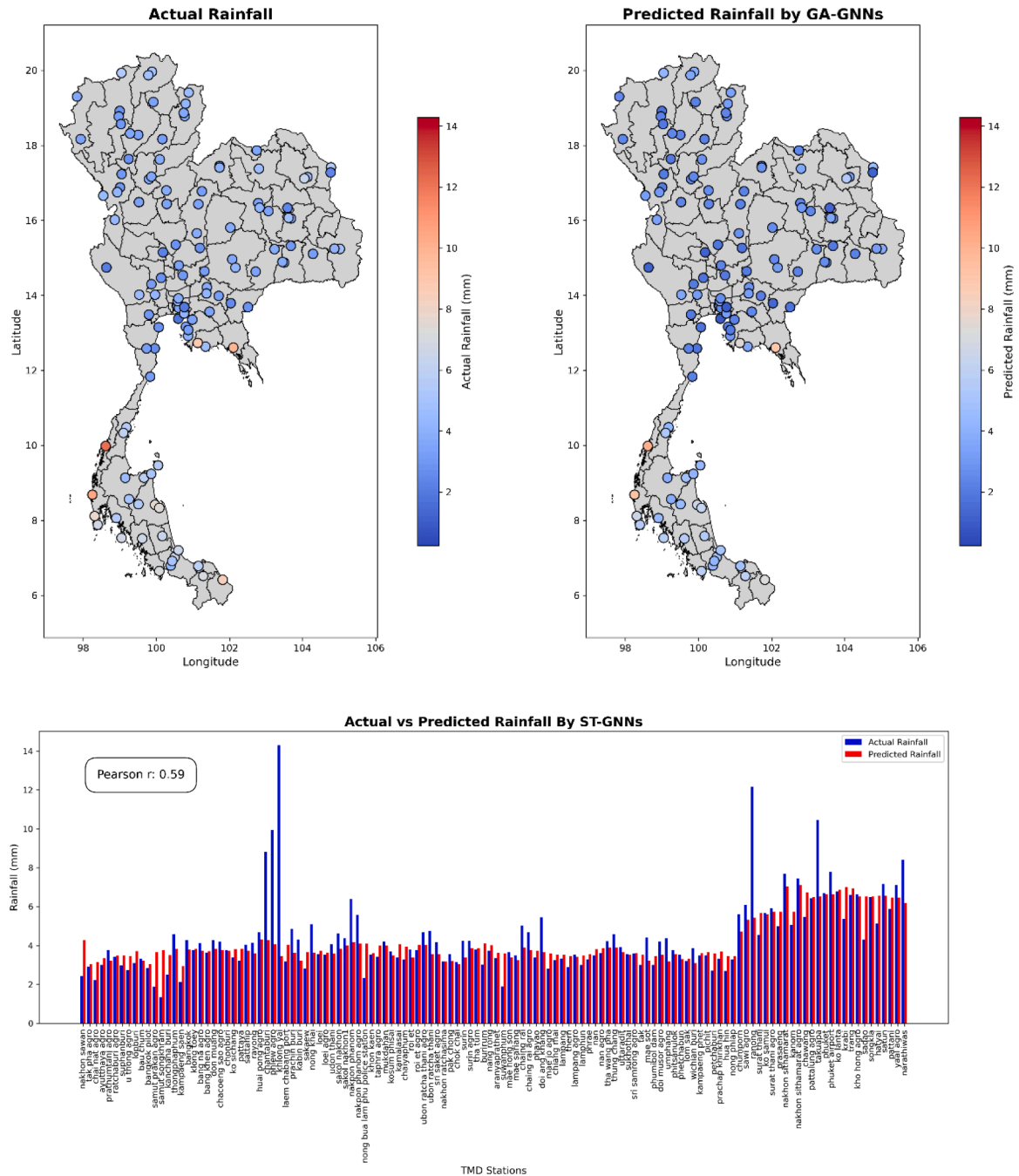


Fig. 11. Actual vs. predicted rainfall by GA-GNNs (a) geospatial comparison, and (b) actual vs. predicted rainfall for each meteorological station with Pearson correlation.

agreement between the observed and the predicted rainfall. A low RMSE of 0.1560 and MAE of 0.3950 are also complementary indicators of its ability to capture complex spatiotemporal rainfall dynamics. On the other hand, the performance of GA-GNN was significantly worse with $R^2 = 0.5800$, $RMSE = 0.8400$, and $MAE = 1.5770$, mainly due to the inability to capture the complex interactions of rainfall patterns.

The SHAP method of analyzing feature importance revealed that WS, RH, and T_{min} were the most significant meteorological factors that influenced ST-GNN performance, which aligns with known fundamental meteorological principles. It is important to note that the positive contribution of WS to model predictions was the greatest (SHAP value = 0.634).

The performance of the classification based on different categories of rainfall intensity further indicated that ST-GNNs achieved almost perfect

accuracy (0.992) and weighted F1 score of 1.00, whereas GA-GNNs registered an accuracy of 0.611 and weighted F1 score of 0.730. These findings show the paramount significance of combining spatial and temporal dynamics in rainfall prediction, especially in conditional climate regions like Thailand.

In spite of the informative results of the current research, a number of limitations are recognized. The imputed data used to fill the time gaps in the historical dataset might have caused model biases, which could have an impact on the performance of the GA-GNN method. Furthermore, the model validation was restricted to Thai meteorological data, thus not allowing generalization to other areas with different climatic regimes. In future studies, it may be useful to increase the spatial extent and add more meteorological variables to the study to improve the strength of the method. Also, the study of hybrid models that combine the ST-GNN

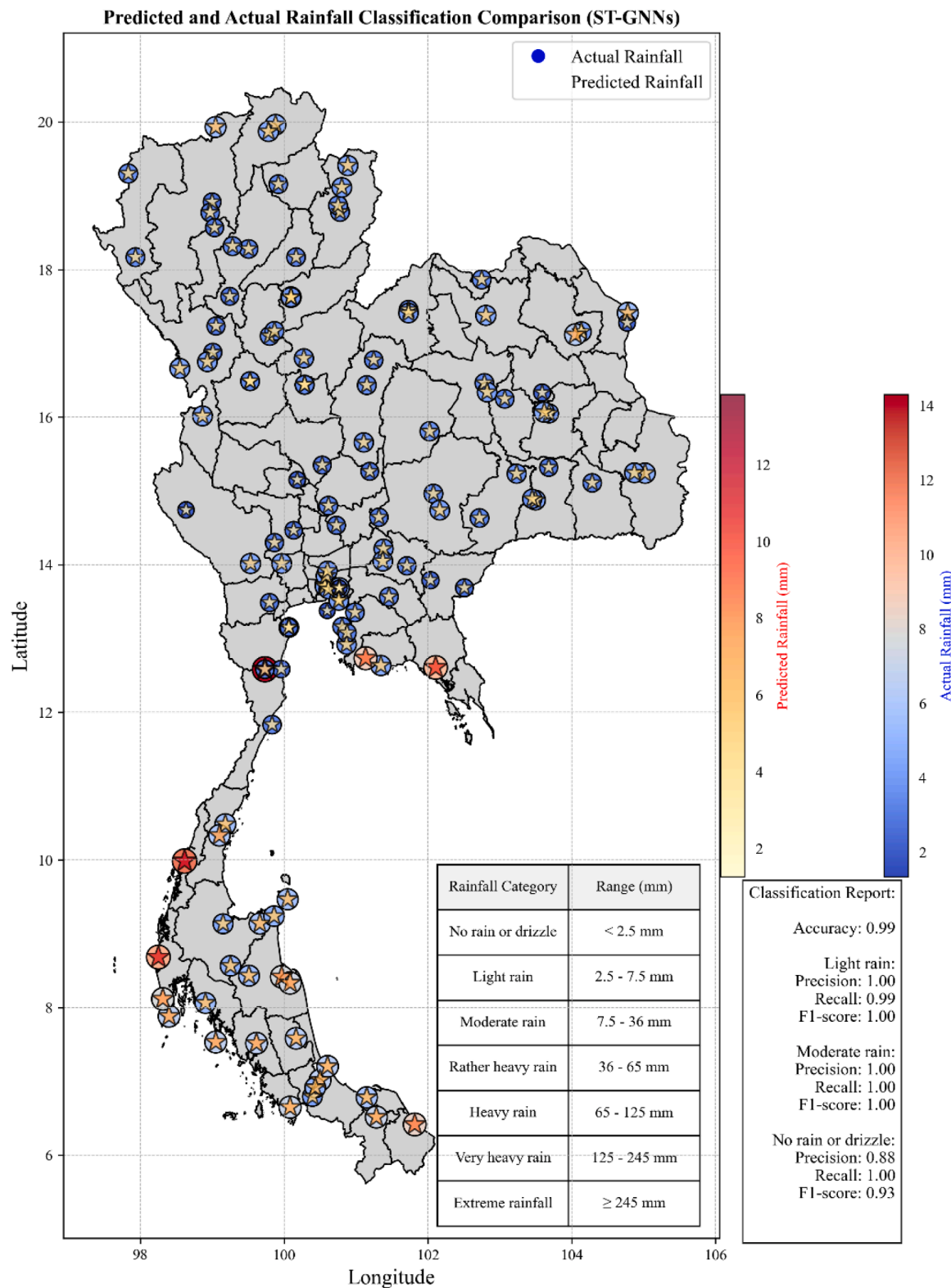


Fig. 12. ST-GNNs performance in the detection of different rainfall categories.

and GA-GNN models has the potential to further increase the predictive accuracy, especially in extreme rainfall events.

Overall, the contributions in the paper make significant contributions to the field of spatiotemporal rainfall forecasting, showing the effectiveness of graph-based neural networks and defining the directions of methodological improvement clearly.

- Abbreviations**
- AI Artificial Intelligence
 - ARIMA Autoregressive Integrated Moving Average
 - CART Classification and Regression Trees
 - CNN Convolutional Neural Networks
 - DL Deep Learning

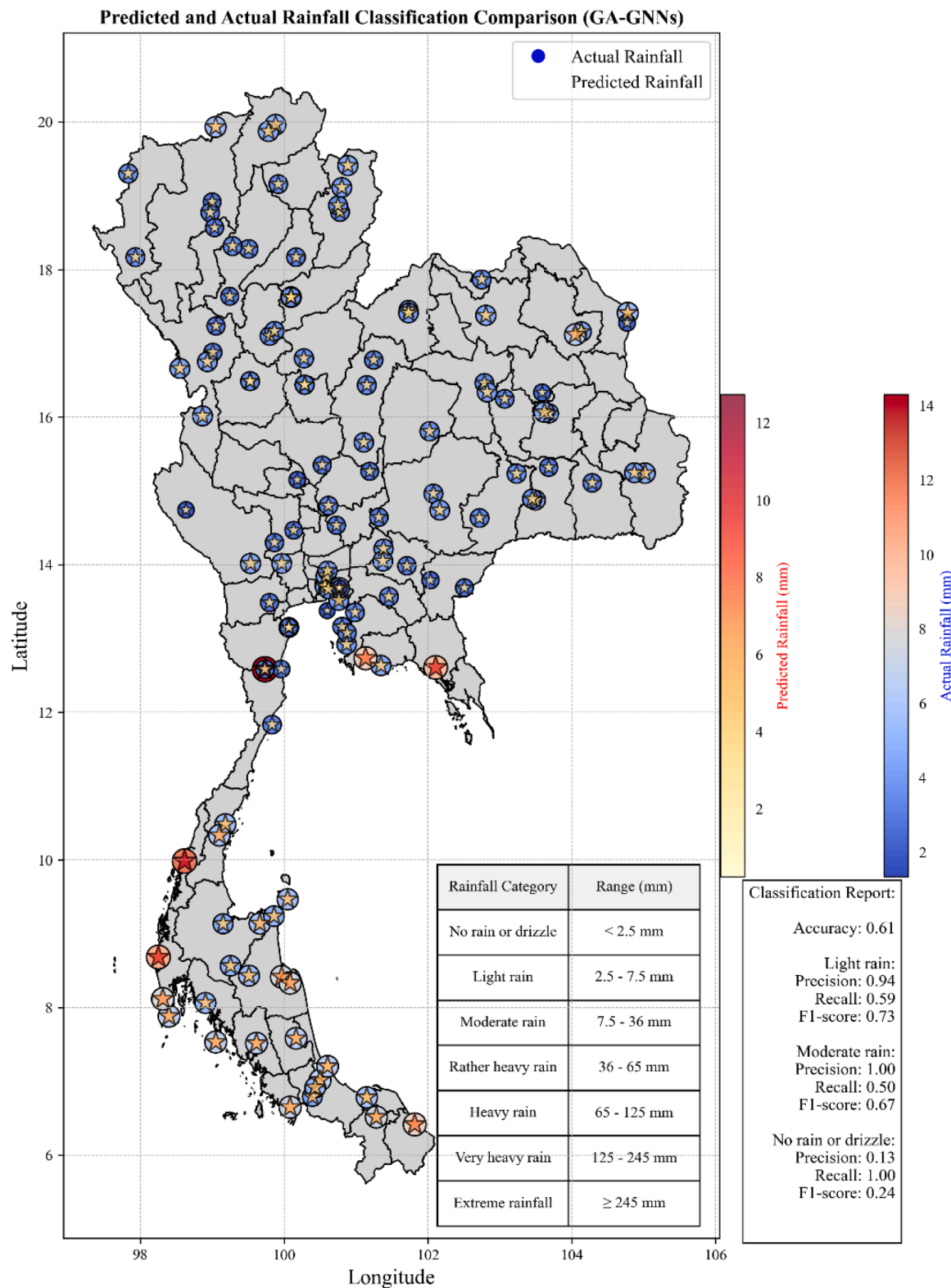


Fig. 13. GA-GNNs performance in the detection of different rainfall categories.

- DTR

Diurnal Temperature Range
- GA-GNNs

Generative Adversarial Graph Neural Networks
- GANs

Generative Adversarial Networks
- GAT

Graph Attention Networks
- GCN

Graph Convolutional Networks
- GNNs

Graph Neural Networks
- GRU

Gated Recurrent Unit
- LSTM

Long Short-Term Memory
- MAE

Mean Absolute Error
- MAPE

Mean Absolute Percentage Error
- ML

Machine Learning
- MLP

Multi-Layer Perceptron

NRMSE	Normalized Root Mean Square Error
NWP	Numerical Weather Prediction
PBIAS	Percent Bias
R ²	Coefficient of Determination
RNNs	Recurrent Neural Networks
RMSE	Root Mean Square Error
RRMSE	Relative Root Mean Square Error
SARIMA	Seasonal Autoregressive Integrated Moving Average
SHAP	Shapley Additive Explanations
SOI	Southern Oscillation Index
SPEI	Standardized Precipitation Evapotranspiration Index
ST-GNNs	Spatio-Temporal Graph Neural Networks
Tmax	Maximum Temperature
Tmin	Minimum Temperature
TMD	Thai Meteorological Department
WS	Wind Speed
WD	Wind Direction
WGAN	Wasserstein Generative Adversarial Network

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CRediT authorship contribution statement

Usa Wannasingha Humphries: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition. **Muhammad Waqas:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shakeel Ahmad:** Writing – review & editing, Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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